

Unit V

5

Input / Output and File Management

5.1 : I/O Devices, Organization of the I/O Function

Q.1 Enlist the characteristics of block and character devices? Explain each with suitable example. [SPPU : May-19, End Sem, Marks 4]

Ans. : Characteristics of block devices :-

- Driver communicates by sending entire blocks of data.
 - Examples for Block Devices : hard disks, USB cameras, Disk-On-Key.
 - Block devices are accessed a block at a time.
 - Operations supported include read(), write(), and seek().
- Character Devices :**
- Character devices are accessed one byte at a time.
 - Supported operations include get() and put().
 - Examples for Character Devices : Serial ports, parallel ports, sounds cards.

Q.2 Explain parameters used for differentiating I/O devices.

Ans. :

1. **Data rate** will change according to the input output devices. There may be differences of several orders of magnitude between the data transfer rates. Data transfer rate of keyboard is the lowest among the entire I/O device.
2. **Application** : Different devices have different use in the system.
3. **Complexity of control** : It will change according to the input-output devices. Printer requires a relatively simple control interface. Disk is much more complex interface.

4. **Unit of transfer** : Data may be transferred as a stream of bytes or characters or in larger blocks.
5. **Data representation** : Different data encoding schemes are used for different devices.
6. **Error conditions** : The nature of errors differ widely from one device to another.

Q.3 List and explain in brief I/O performing techniques (at least three). [SPPU : Dec-17, End Sem, Marks 12]

OR Discuss at least two techniques for performing I/O.

Ans. : I/O performing techniques are DMA, Direct I/O with polling and Interrupt driven I/O.

1. DMA :

- Direct Memory Access (DMA) is an I/O technique that allows a control unit to directly access main memory. This means that once reading or writing has begun the remainder of the data can be transferred to and from memory without processor (CPU) intervention.

• Fig. Q.3.1 shows Operation of the DMA.

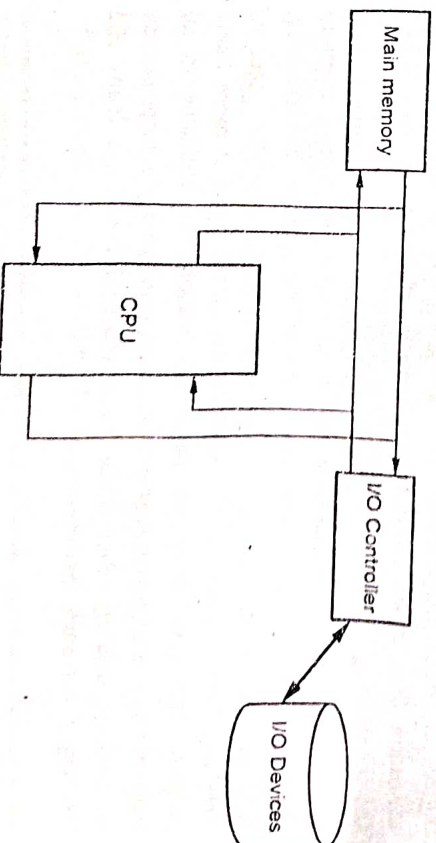


Fig. Q.3.1

- Without DMA, the processor is responsible for the physical movement of data between main memory and a device.
- A special control unit may be provided to allow transfer of a block of data directly between an external device and the main memory, without continuous intervention by the processor. This approach is called Direct Memory Access.
- DMA controller is programmed by CPU and its registers are set to perform required operations. It issues a command to a DMA controller for reading data from the secondary storage disk and stores it in the internal buffer. It verifies the checksum. If disk controller buffer contains valid data then DMA operation starts.
- 2. **Direct I/O with polling :** CPU is responsible for data transfer. It transfers data between primary memory and device controller. Following are the steps for polling :
 - a) Application process issues a read operation.
 - b) Device driver check the status of device. If device is busy, the driver waits for it to become idle.
 - c) Driver stores an input command into the controller command register. It starts the device.
 - d) Driver continuously read the status register while waiting for the device to complete its operations.
 - e) Driver copies the content of the controller data register into the user process space.
- 3. **Interrupt driven I/O :** When interrupt I/O is used, the CPU is free to ignore the I/O module until the interrupt signal is received. The only added overhead is processing a single interrupt when the I/O operation completes.

5.2 : I/O Buffering

Q.4 What is I/O buffering? Explain input and output buffering.

Ans. : • Buffering is the technique by which the device manager can keep slower I/O devices busy during times when a process is not requiring I/O operations. Input/output buffering is a mechanism that

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- improves the throughput of input and output operations. It is implemented directly in hardware and the corresponding drivers.
- Input buffering is the technique of having the input device read information into the primary memory before the process requests it.
 - Output buffering is the technique of saving information in memory and then writing it to the device while the process continues execution.

Q.5 Explain in brief different I/O buffering techniques.

[SPPU : Dec.-17, End Sem, Marks 4]

OR Explain I/O buffering mechanism.

[SPPU : May-18, Dec.-18,19, End Sem, Marks 4]

Ans. : Types of I/O buffering schemes are

1. Single buffering
 2. Double buffering
 3. Circular buffering
 4. No buffering
- 1. Single Buffering**
- Operating system assigns a buffer in the system portion of main memory. Fig. Q.5.1 shows single buffering.

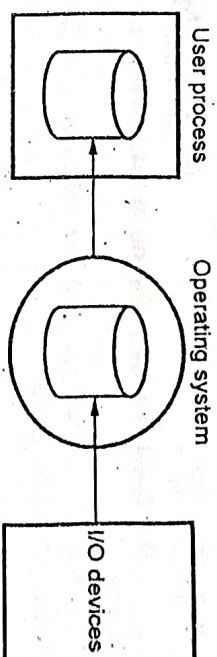


Fig. Q.5.1

2. Double Buffering

- There are two buffers in the system.
- One buffer is for the driver or controller to store data while waiting for it to be retrieved by higher level of the hierarchy.
- Other buffer is to store data from the lower level module.
- Double buffering is also called buffer swapping.
- Fig. Q.5.2 shows the hardware and software double buffering for bytes.

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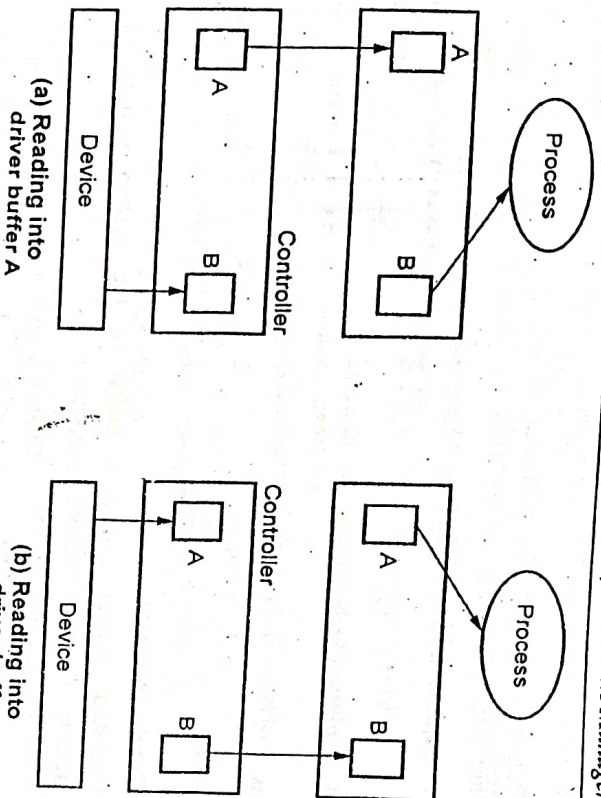


Fig. Q.5.2 Double buffering in the driver.

Circular Buffer

- When more than two buffers are used, the collection of buffers is itself referred to as a circular buffer.
- In this, the producer cannot pass the consumer because it would overwrite buffers before they had been consumed.
- The producer can only fill up to buffer $j-1$ while data in buffer j is waiting to be consumed.
- Fig. Q.5.3 shows circular buffering.

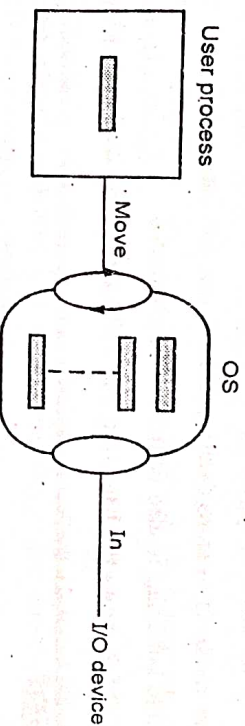


Fig. Q.5.3 Circular buffering

No Buffering : there is a buffer between user process, operating system and I/O device.

5.3 : Disk Scheduling

Q.6 Define following terms : i) Seek time ii) Rotational latency

[SPPU : Dec-17, End Sem, Marks

Ans. :

1. **Seek time :** The time it takes to position the head at the track known as seek time.

Seek time = Number of track traversed $\times$ Disk drive constant + Start time

Seek time do not apply to device with fixed read/write heads.

2. **Rotational latency :** The time it takes for the beginning of the sector to reach the head is known as rotational delay. It is known as seek time.

Q.7 Explain SSTF and LOOK disk scheduling algorithm.

Ans. : Shortest Seek Time First

- SSTF select the next request at the one requiring the minimum seek time from the current position.

- In Shortest-Seek-Time-First (SSTF) scheduling priority is given to those processes which need the shortest seek, even if these requests are not the first ones in the queue. It means that all requests nearer to the current head positions are serviced together before moving head to distant tracks.

- Select the disk I/O request that requires the least movement of the disk arm from its current position and always choose the minimum seek time.

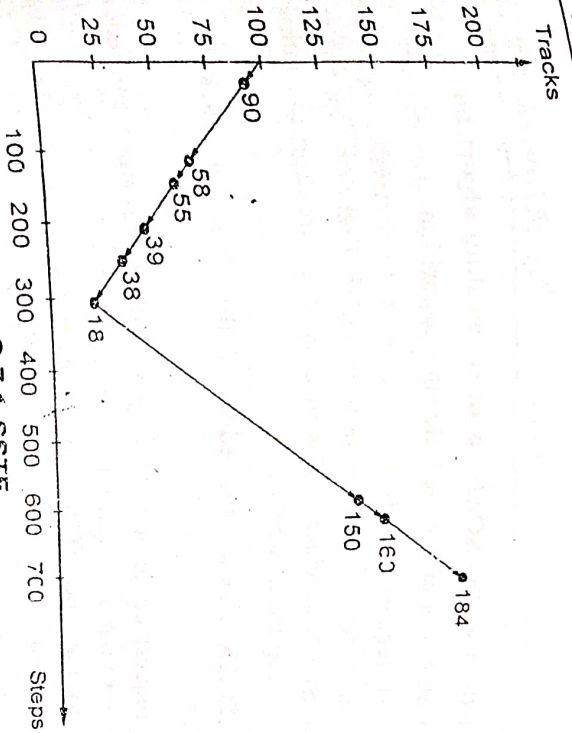


Fig. Q.7.1 SSTF

Next track accessed	Number of track traversed
90	10
58	32
55	03
39	16
38	01
18	20
150	132
160	10
184	24

$$\text{Average seek length} = \frac{10 + 32 + 3 + 16 + 1 + 20 + 132 + 10 + 24}{9} = \frac{248}{9} = 27.55$$

- The only tricky part is if there are two jobs with the same value, you could just use a random number to effectively flip a coin and pick one.

- SSTF does not ensure fairness and can lead to indefinite postponement because its seek pattern tends to be highly localized.
- Under heavy load, SSTF can prevent distant requests from ever being serviced. This phenomenon is known as starvation. SSTF scheduling essentially a form of shortest job first scheduling. SSTF scheduling algorithm are not very popular because of two reasons.

1. Starvation possibly exists.
2. It increases higher overheads.

Advantages :

1. Throughput is better than FCFS.
2. SSTF minimize the response time.
3. Less number of head movements.

Disadvantages :

1. One major issue is STARVATION.
2. Localize under heavy load.

LOOK Scheduling

- Start the head moving in one direction. Satisfy the request for the closest track in that direction when there is no more request in the direction, the head is traveling, reverse direction and repeat. This algorithm is similar to SCAN, but unlike SCAN, the head does not unnecessarily travel to the innermost and outermost track on each circuit.

- Fig. Q.7.2 shows the look scheduling algorithm. For example, the disk request queue contains a set of references for blocks on tracks 76, 124, 17, 269, 201, 29, 137 and 12.

Current head position = 76

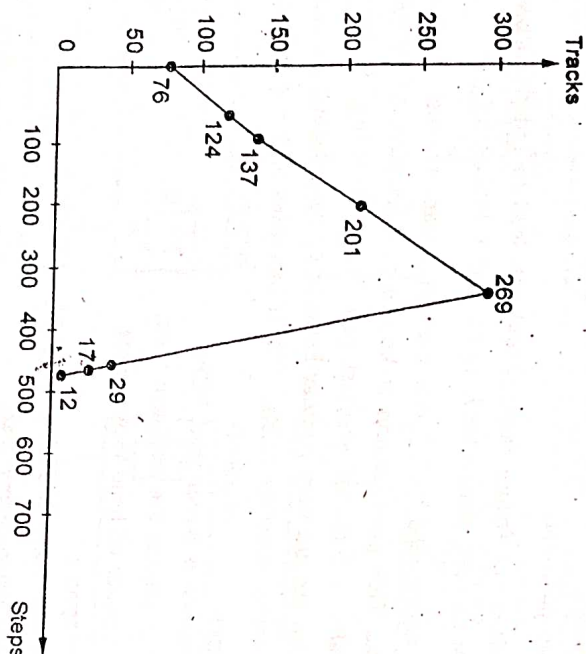


Fig. Q.7.2 LOOK scheduling

Next track accessed	Number of track traversed
124	48
137	13
201	64
269	68
29	240
17	12
12	5

$$\text{Average seek length} = \frac{48 + 13 + 64 + 68 + 240 + 12 + 5}{7} = \frac{450}{7} = 64.28$$

Q.8 Briefly describe SCAN disk scheduling algorithm.

Ans. : • SCAN is also called elevator algorithm.

- The next request scheduled is closest to current request but in particular direction. All requests in other direction are put at the end of the list. SCAN services tracks in only one direction i.e. either increasing or decreasing track number.
- When SCAN reaches the edge of the disk (or track 0), it reverses direction.
- If any request comes on its way it will be serviced immediately, with request arriving just behind the head will have to wait until disk head moves to the end of the disk, reverses direction and returns before being serviced.

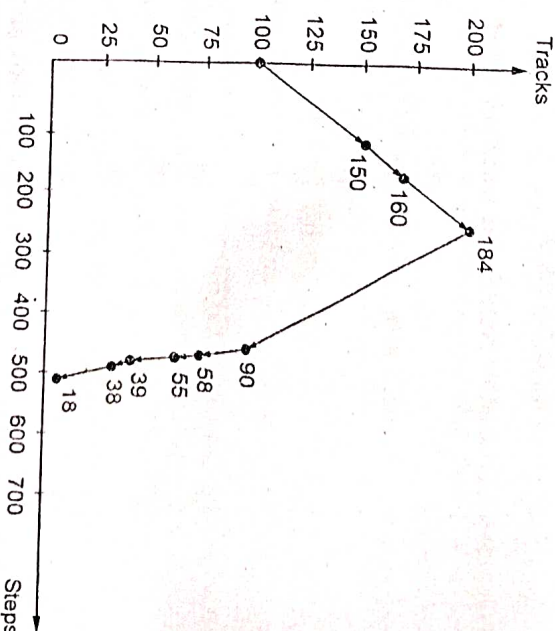


Fig. Q.8.1 SCAN

- SCAN is called elevator because an elevator continues in one direction servicing requests before reversing direction.
- Consider the example in which the requested tracks in the order received are : 55, 58, 39, 18, 90, 160, 150, 38, and 184. Starting track

at 100 and total number of track is 200. Head is moving towards increasing track number.
Current head position = 100
Head is moving towards increasing number of track.

Next track accessed	Number of track traversed
150	50
160	10
184	24
90	94
58	32
55	03
39	16
38	01
18	20

$$\text{Average seek length} = \frac{50 + 10 + 24 + 94 + 32 + 3 + 16 + 01 + 20}{9} = \frac{250}{9} = 27.77$$

Advantages :

1. SCAN eliminates starvation.
2. Throughput similar to SSTF.

Disadvantages :

1. Increase overhead.
2. Needs directional bit.

Q.9 A disk drive has 200 cylinders, numbered 0-199. The drive is currently serving the request at cylinder 53. The queue of pending requests in FIFO order is 98, 183, 37, 122, 14, 124, 65, 67. Starting from the current head position what is the total distance that disk arm moves to satisfy all the pending requests for the following disk scheduling algorithms. i) FCFS ii) SCAN

iii) C-LOOK iv) SSTF [SPPU : Dec-17, End sem, Marks 12]

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Ans. : i) FCFS :

Head movement = 53 → 98 → 183 → 37 → 122 → 14 → 124 → 65 → 67
Total head movement = 640

ii) SCAN :

Head movement = 53 → 37 → 14 → 0 → 65 → 67 → 98 → 122 → 124 → 183
Total head movement = 208

iii) C-LOOK :

Head movement = 53 → 65 → 67 → 98 → 122 → 124 → 183 → 14 → 37
Total head movement = 236

iv) SSTF :

Head movement = 53 → 65 → 67 → 37 → 14 → 98 → 122 → 124 → 183
Total head movement = 236

Q.10 For the given sequence of disk request, determine the total distance travelled by disk head in satisfying the entire request for FCFS, C-SCAN and SSTF algorithms. Initial head position is 120 and total number of cylinders in the disk is 200.

120, 130, 180, 150, 25, 10, 105, 90 [SPPU : May-17, End sem, Marks 12]

Ans. : i) FCFS :

Head movement = 120 → 130 → 180 → 150 → 25 → 10 → 105 → 90
Total head movement = 340

ii) C-SCAN :

Head movement = 120 → 130 → 150 → 180 → 200 → 1 → 10 → 25 → 90 → 105
Total head movement = 383

iii) SSTF :

Head movement = 120 → 130 → 150 → 180 → 105 → 90 → 25 → 10
Total head movement = 230

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Q.11 Is disk scheduling, other than FCFS useful in a single user environment. Explain your answer. [SPPU : Dec-16, End Sem, Marks 6]

Ans. : • Yes, disk scheduling other than FCFS scheduling can be useful in a single-user environment.

• SSTF is useful when the data sets are near each other. In the elevator scheduling, we saw that the effects are optimal because it is more efficient than FCFS but unlike SSTF, there are no starvation.

• In a single-user environment, the I/O queue usually is empty. Requests generally arrive from a single process for one block or for a sequence of consecutive blocks.

• In these cases, FCFS is an economical method of disk scheduling. But LOOK is nearly as easy to program and will give much better performance when multiple processes are performing concurrent I/O, such as when a Web browser retrieves data in the background while the operating system is paging and another application is active in the foreground.

Q.12 A disk drive has 200 cylinders, numbered 0 - 199. The drive is currently serving the request at cylinder 63. The queue of pending requests in FIFO order is 27, 129, 110, 186, 147, 41, 10, 64, 120. Starting from the current head position what is the total distance that disk arm moves to satisfy all the pending requests for the following disk scheduling algorithms. i) FCFS ii) C-SCAN iii) C-LOOK iv) SSTF [SPPU : May-18, Dec-18, 19, End Sem, Marks 12]

Ans. : Currently serving the request at cylinder 63.

FCFS	C-SCAN	C-LOOK	SSTF
Next Request	Next Request	Next Request	Next Request
Difference in head movement	Difference in head movement	Difference in head movement	Difference in head movement
27	64	1	64
36	1	64	1
129	110	46	41
102	46	110	23
110	120	10	27
19	10	120	14
186	129	9	17
76	9	129	10

147	39	147	18	147	18	110	100
41	106	186	39	186	39	120	10
10	31	0	186	10	176	129	9
64	54	10	10	27	17	147	18
120	56	27	17	41	13	186	39
		41	14				

Total head movement in FCFS = 519

Total head movement in C-SCAN = 349

Total head movement in C-LOOK = 329

Total head movement in SSTF = 231

Q.13 Suppose that a disk drive has 400 cylinders, numbered 0 to 399. The drive is currently serving a request at cylinder 160. The queue of pending requests, in FIFO order, is 370, 30, 390, 130, 310, 170, 340, 180. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk scheduling algorithms ?

i) FCFS ii) SSTF iii) SCAN iv) LOOK
(Assume that the head is initially moving towards track number 399).

Ans. : Disk scheduling algorithm : Starting head position at 160

FCFS	SSTF	SCAN	LOOK
Next request	Next request	Next request	Next request
Difference in head movement	Difference in head movement	Difference in head movement	Difference in head movement
370	210	170	170
30	340	180	180
390	360	130	130
130	260	30	310
310	180	310	340
		280	370
		30	370

170	240	340	30	390	20	390	20
340	170	370	30	399	9	130	260
180	160	390	20	130	269	30	100
Total	1920	Total	530	30	100	Total	590
		Total		608			

5.4 : Overview of File Management

Q.14 Discuss the following with respect to file system : file attributes.

Ans. : File Attributes

- One of the characteristics of file is a set of file attributes that give the operating system more information about the file and how it is intended to be used. For human users convenience file is given to the file. File attributes are varies from system to system. File attributes are as follows :
 1. Name
 2. Identifier
 3. Type
 4. Location/place
 5. Size
 6. Protection
 7. Date and time
- File name is in human readable. User can perform any operation on file using its name.
- When file is created by user, operating system assigns unique identification number to each file. OS uses this identifier for its operation.
- Type information is required for systems that support different types of files.
- Location/place information is pointer to a device and actual location of files on the device. The information needed to locate the file on disk is kept in memory.
- Size is measured in bits or bytes. It gives idea about current size of the file.
- Protection : This information is stored on the per-process table so the operating system can allow or deny subsequent requests. Attributes like protection, password, creator and owner provides protection to a file.

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- Date and time : This information is related to creation and modification of files. Protection, security and monitoring purposes this data is used by operating system.

Q.15 Discuss the following with respect to file system : file operation

Ans. : • File operations are simply those things that user can perform on a file. For example, user can create a file, save a file, open a file, read a file and modify a file. There are many different types of file operations supported by operating system.

- Operating system can provides system call for performing various operations on the file. Six basic file operations are creating, write, read, delete, repositioning and truncating.
- Following are the some list of file operation :
 1. Create
 2. Write
 3. Close
 4. Read
 5. Delete
 6. Repositioning
- All these operations require that the directory structure be first searched for the target file.
- File creation : Any time user can create a file. File is also created without data. The steps for creating a file.
 - a. Space : File system must provide a space for the file.
 - b. New file entry is made in the directory.
- Write : To perform write operation on the file, file name and data is required. It updates a file by adding new data and changes some content of the file. For writing into a file, operating system searches the directory for particular file. Write pointer is kept at the location where the writing starts. Write pointer position is at middle of the file or end of the file. Write pointer is updated when writing to file is completed.
- Close : Process is not used a file after closing. Process can not perform any operation on file. Operating system remove it's entry from open file table.
- Read : A process reads all or some portion of the data in a file. Read system call is used for reading a file. For reading a file, name and

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position of the memory is specified. Read pointer is kept into the file for next read operation. Read pointer is updated after completion of the read system call. Information about read pointer is stored in *per process current file position pointer*.

- **Delete :** Operating system remove the file from file structure. For deleting a file, directory is searched with particular name and file is deleted. Deleting means making space free from memory and disk. Operating system also removes the directory entry. This free space is used by another file.
- **Repositioning :** The directory is searched for the proper entry, and the current-file-position pointer is repositioned to a given value. Repositioning within a file need not involve any actual I/O. This file operation is also known as a file seeks.
- Other file operations are open, append, seek, get and set attributes and rename.
- **Open :** Before performing any operation on file, it is necessary to open file in required mode.
- **Append :** Append is related to write operation of file. Append only add data to the end of the file.
- **Get and set attributes :** Get attribute is used before performing operation on file. When user request a file for any operation, get attributes is executed by the process. Depending upon the attributes, user can allow or deny the operation. User can set the attributes and it changed after creation of file.
- **Rename :** User can change the name of existing file.

Q.16 What is the objective of file management system ?

Ans. : Objective of file management system

1. Provide I/O support the multiple users.
2. Provide I/O support for a variety of secondary storage device.
3. Support for data management
4. It minimizes the data lost.
5. Optimize the system performance.

Q.17 Draw and explain file system architecture.

Ans. : File System Architecture

• Fig. Q.17.1 shows file system architecture.

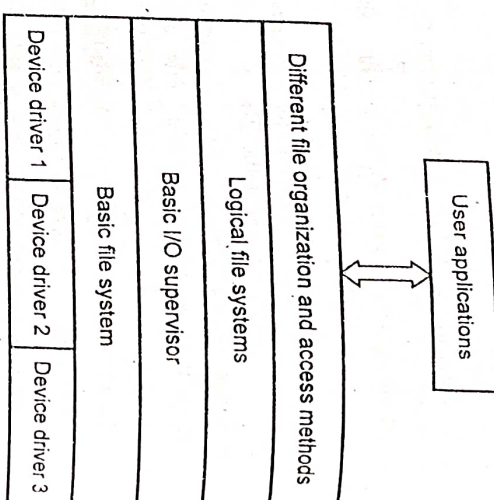


Fig. Q.17.1 File system architecture

- Device driver**
- Device driver directly communicate with peripheral device and controllers.
- Basic file system**
- It is responsible for starting physical I/O operations on the device.
 - It uses specific device driver.
 - It is primary interface used by system for communicating with our world.
- Basic I/O Supervisor**
- It exchange block of data with secondary storage device.
 - It is part of the operating system.
- Basic I/O Supervisor**
- It is responsible for initialize all file I/O and termination.
 - It is also part of operating systems.
 - It select the device on which file I/O is to be performed.
 - It allocates I/O buffer.

5.5 : File Organization and Access

Q.18 Explain different file organization techniques ?

[SPPU : Dec-16, End Sem, Marks 8]

Ans : • Different file organization techniques are Sequential Access, direct access, Indexed sequential access.

Sequential Access Method

- Sequential access method is simple method. The information in a file is accessed sequentially one record after another.
- In this method, a process could read all the records in a file in order, starting at the beginning. It can not skip any records and can not read out of order. Fig. Q.18.1 shows sequential access method.

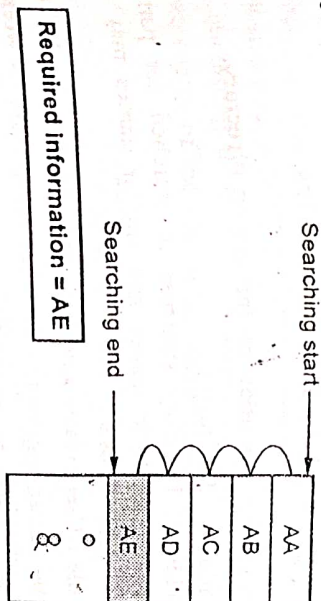


Fig. Q.18.1 Sequential file access

- Batch application uses sequential files. A byte stream file uses sequential access method.
 - Example : Compiler and editor usually access files in this method. Transaction file is also example of sequential access method.
 - Sequential file organization only method easily stored on tape and hard disk. Sequential access is best suited where most of the records in a file are to be processed.
- Disadvantages :**
- It provides poor performances.
 - More efficient search technique is required.

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Direct Access Method

- It is also called random access method. This access allows a user to position the read/write mark before reading or writing. Fig. Q.18.2 shows direct access method.

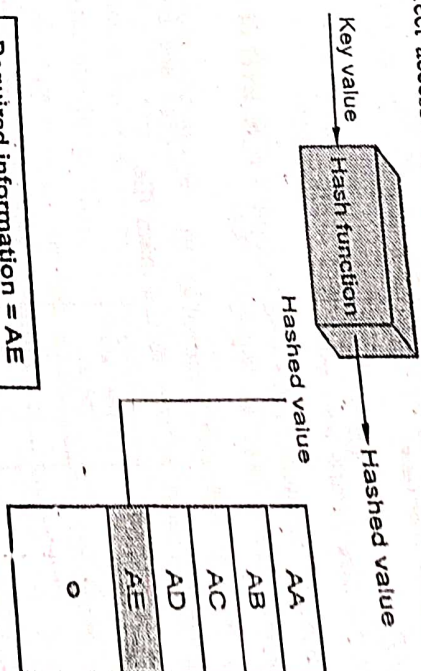


Fig. Q.18.2 Direct access method

- Direct access file method provides, accessing the records directly. Direct access method is based on the hard disk that is a direct access device. It allows random access of any file block.
 - Each record has its own address on the file with the help of which it can be directly accessed for reading or writing. This feature is used by editors. An editors need to randomly access the contents of the file.
 - There is no restriction on the order of reading or writing for a direct access file. Operating system support is not required for direct access file.
- Disadvantages :**
1. Poor utilization of input-output device.

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2. Consumes CPU time for record address calculation.

Indexed Sequential Access

- It is modified method of direct access method. This method is combination of direct and sequential access method.
- In simple indexed sequential method, simple single level of indexing is used. Sequential file is used as index.

- Fig. Q.18.3 shows indexed sequential file. Indexed sequential access contains key field and a pointer to the main file.

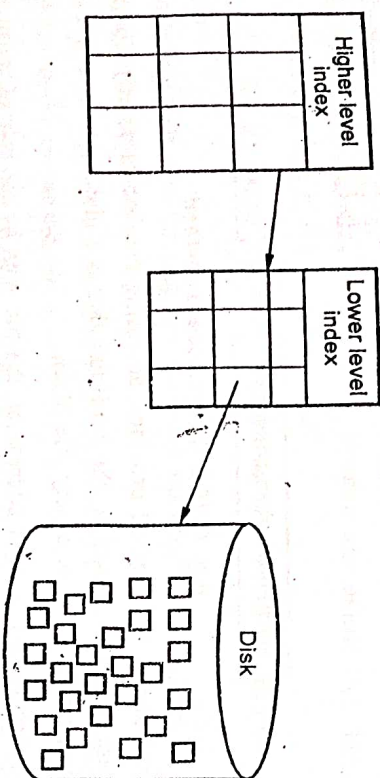


Fig. Q.18.3 Indexed sequential access

- Indexed sequential files are store records in the order that they are written to the disk. Records may be retrieved in sequential order or in random order using an index to represent the record number in the file.
- If file size is large, larger index is required and it contains large number of entries. CPU takes time to search into the index. Higher level index is used to reduce the search time. An entry in the higher level index points to a section of the index. This higher index contains the records. This index is searched to find the section of the disk that may contains a required record.
- File system maintains an overflow file. It adds new records to an overflow file. The record in the main file that immediately precedes the new record in logical sequence is updated to contain pointer to the new record in the overflow file.

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- The overflow file is merged with the main file during a batch up. The multiple indexes for the same key field can be set up to improve efficiency.
- The lowest level of index file is treated as a sequential file and a higher level index file is created for that file. Higher index file would contain pointers to lower index files, which would point to the actual items.

5.6 : File Directories

Q.19 Explain various operation performed on directory.

Ans. : Operation on directory :

1. File searching : Directory structure is searched for particular entry. File uses symbolic names and similar names may indicate relationship between files.
2. Create a file : User creates new files and added to the directory.
3. Delete a file : User remove the file after completing its work on file.
4. Rename a file : User change the file name if file content is change.
5. Listing of directory : Listing of files in the directory for some MS-DOS and windows uses "dir" command and Linux/UNIX uses "ls" command for this purposes.

Q.20 Write a short note on : Directory structure.

Ans. : ISPPU : May-18, Dec-18,19, Eng Sem, Marks 4

- Directories are basically symbol tables of files. A single file directory can contain a list of all files in a system. A directory contains information about the files, including attributes, location and ownership.
- All information about files is kept in the directory structure. It is store on the secondary storage device.
- Single level directory : Single level directory is a simple structure and there are no sub-directories. It is also called flat directory structure.
- In single level directory structure, no two files can have the same name.
- Two level directory : Two-level directory structure contains master file directory at the top level then user file directory at the second level. Actual user files are at the third level.

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• File system maintains a master block for each user. Master block has one entry for each user. User directory address is stored in the master block. Each user has a private directory.

• **Tree structured directory :** In this structure, directory itself is a file. A directory and sub directory contains a set of files. Internal format is same for all directories. Commonly used directory structure is tree structure. Tree has a root directory. All files in disk have a unique path name.

Q.21 Explain acyclic graph tree.

Ans. : • Acyclic graph directory solve the problem of tree structure directory. It allows sharing the directory in between two users. At a time more than one places, shared directory or file will exist in the file system. Fig. 5.8.4 shows acyclic graph directory. Links can be used to construct acyclic graph structure. Acyclic graph is graph with no cycles.

• It is very interesting to note that a shared directory or file is not the same as two copies of the file. When there are two copies of files, each

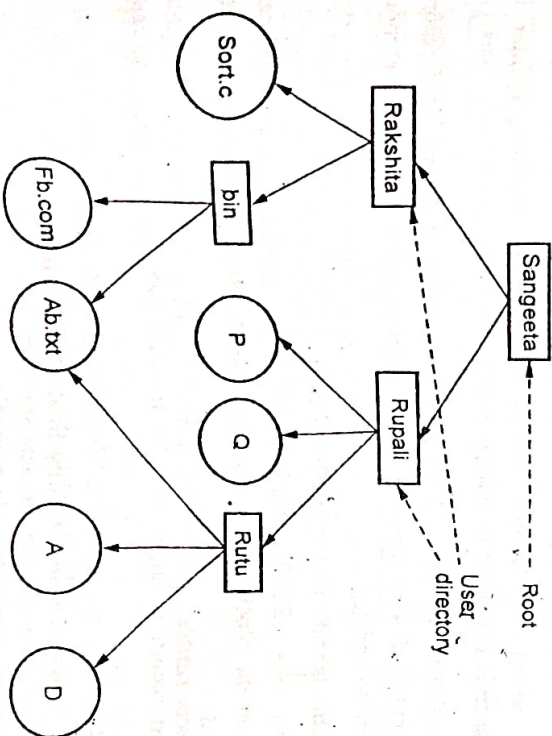


Fig. Q.21.1 Acyclic graph directory

user can view the copy rather than the original, but if one user changes the file content, the changes will not appear in the other's copy.

• Only one original file exists for shared copy. Any changes made by one user are immediately visible to the other user. When user create file in shared directory, it automatically appear in all the shared subdirectories.

5.7 : File Sharing

Q.22 Write a short note on : File sharing.

[SPPU : May-18, Dec-18,19, End Sem, Marks 4]

Ans. : • Multiprogramming and multitasking operating system support the multiple user. These users perform various operations on the files. Files are shared between number of users. File protection, naming and sharing is issue for file sharing.

• Directory structure may allows files to be shared by users. Sharing must be done through a protection scheme.

• In single user system, operating system maintains attributes for all files and directory. According to the attributes, operating system allows to operate the files for users. But on the multi-user system, more attributes are needed. An attributes like owner, group and other user are required. These are the access permission for files and directory.

• Access permissions are of three types : Read, Write and execute.

1. An "r" indicates read permissions for a file or directory.
2. A "w" indicates write permissions.
3. An "x" indicates execution permissions. if it is a file and search permissions if it is a directory.
4. Owner is the file owner who created a file. Owner have all rights (read, write and execute).
5. The groups of other user have some limited access permission. Mostly read and execute permission.
6. Last categories of user have only executed permission.

• Example : In UNIX operating system, file owner can perform all operation on a file. Group member can execute one subset of those

operations and other than the group user can execute another subset of operations.

- When a user requests an operation on a file, the user ID can be compared with the owner attribute to determine if the requesting user is the owner of the file. Likewise, the group IDs can be compared.

5.8 : Record Blocking

Q.23 List and briefly define three blocking methods.

Ans. : • File is said to employ blocking of records if a physical record contains more than one logical record. Records are the term used to refer to a number of related values or items. Each value or item is stored in a field of a specific data type. Records may be of either fixed or variable lengths.

- Blocks are of fixed length. Disks are divided into concentric circular tracks on each disk surface. A track is divided into blocks. Whole blocks are transferred between disk and main memory for processing.
- Blocking is refers to storing a number of records in one block on the disk. The blocking factor refers to the number of records per block. There may be empty space in a block if an integral number of records do not fit in one block.
- Spanned records are referring to records that exceed the size of one or more blocks and hence span a number of blocks.
- Record blocking are of three types :
 1. Fixed blocking.
 2. Variable length spanned blocking.
 3. Variable length unspanned blocking
- Fixed blocking uses fixed length records. Fig. Q.23.1 shows fixed blocking.

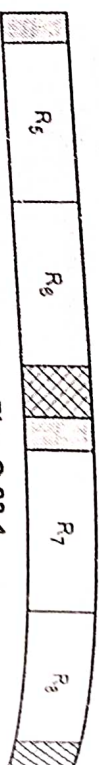
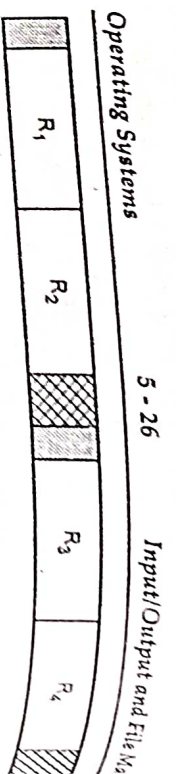


Fig. Q.23.1

- After allocating data, it creates unused space at the end of the block. It is called internal fragmentation. Internal fragmentation is the block.
- Fixed blocking is common for sequential files with fixed-length.
- 2. Variable length spanned blocking
 - In this blocking method, record sizes are not same. It uses length records. These records are packed into blocks without any space.
 - Some record may consume two blocks. In this situation, the record is passed from one block to another block.
 - It is efficient in length and efficiency of storage. It doesn't waste space, but it is more complicated to implement. So, the files are difficult to update.
- 3. Variable length unspanned blocking
 - It also uses variable length records. But records are not packed between the blocks. It creates larger waste space as compared to the method.
 - If next record is larger than the spanned size, then it creates a new space.

5.9 : Secondary Storage Management

Q.24 Explain contiguous file allocation.

Ans. : • When user creates a file, system allocates a set of contiguous blocks on disk. This is a pre-allocation method that uses pre-allocated variable size. Contiguous allocation is simple method to implement.

only search free list of correct number of consecutive blocks and mark them as used block.

- Disk address is a linear ordering on the disk. Because of linear ordering, accessing block $b + 1$ after block b normally requires no head movement. Here head movement is only one track. Fig. Q.24.1 shows contiguous allocation.

File allocation table

File name	Start block number	Block length
File 1	1	3
File 2	18	2
File 4	6	6
File 5	20	1
File 6	25	3

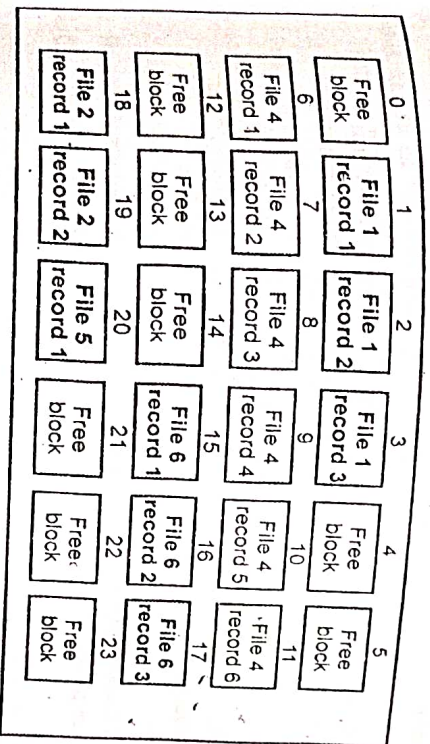


Fig. Q.24.1 Contiguous allocation

- Contiguous allocation of a file is defined by the disk address and the length of the first block.
- If the file size is "n" blocks and starting location is "L", then it occupies blocks L, L+1, L+2, L+3, ..., L+(n-1). The directory entry for each file indicates the address of the starting block and the length of the area allocated for this file.

Decode

- Sequential and random access can be supported by contiguous allocation.
- It is easy to retrieve single block. To improve the I/O performance of sequential processing, multiple blocks can be brought in one at a time. It supports sequential access very well because files entire data is stored in adjacent block. This method also supports random access.
- Contiguous allocation also suffers from external fragmentation. Small free disk spaces are created after allocation free block and deleting files. External fragmentation means there will require free space available but that is not contiguous. To solve the problem of external fragmentation, compaction method is used.
- One more problem is that how to calculate the space needed for a file. It is difficult to estimate the required space.

Q.25 Describe indexed allocation with its advantages and disadvantages.

Ans.: • Indexed allocation method solves the problem of both contiguous and linked allocation. It uses concept of index block. Index block stores the entire pointer in one location. But the index block will occupy some space and thus could be considered as an overhead of the method.

- OS keeps a list of free blocks. It allocates an array to hold pointers to all the blocks used by the file. It allocates blocks only on demand. Fig. Q.25.1 shows indexed allocation. (Refer Fig. Q.25.1 on next page)

- In indexed allocation, each file maintains its own index block. It contains an array of disk sector of addresses. For example: The n^{th} entry in the index block points to the n^{th} sector of the file. The directory contains the address of the index block of a file. To read the n^{th} sector of the file, the pointer in the n^{th} index block entry is read to find the desired sector.

- It supports direct access and without suffering from external fragmentation. Any free block anywhere on the disk may satisfy a request for more space.

Decode

File name	Index block number
File 1	10
File 4	24

0	1	2	3	4	5	6	7
					File 1		
8	9	10	11	12	13	14	15
	File 4	5	14			File 1	
		31					
16	17	18	19	20	21	22	23
				File 4			File 4
24	25	26	27	28	29	30	31
9					File 4		File 1
20							
23							
29							

Secondary storage device

Fig. Q.25.1 Indexed allocation

• Indexed allocation does suffer from wasted space. The pointer overhead of the index block is generally greater than the pointer overhead of linked allocation.

Advantages :

1. It supports sequential and direct access.
2. No external fragmentation.
3. Faster than other two methods.
4. It supports fixed and variable size blocks.

Disadvantages :

1. Indexed allocation does suffer wasted space.
2. Pointer overhead is generally greater.

Q.26 Compare contiguous, linked and indexed file allocation methods.

Ans. :

Sr. No.	Contiguous	Linked	Indexed
1.	Allocate each file to contiguous blocks on disk	Allocate linked-list of fixed-sized blocks	Allocate fixed-size blocks for each file
2.	It suffers from external fragmentation.	No external fragmentation	No external fragmentation
3.	Pre-allocation is required	Pre-allocation is possible	Pre-allocation is possible
4.	It support variable size portions.	It support fixed size portions.	It supports fixed and variable size portions.
5.	Simple to calculate random addresses	Cannot calculate random addresses without reading previous blocks	Supports random access
6.	Allocation frequency is once	Allocation frequency is low to high	Allocation frequency is medium
7.	No pointer overhead	Space required for pointer	Pointer overhead is generally greater